

the IAR Embedded Workbench Kickstart has a 4k or 8k code limitation depending on the processor used. The Launchpad board also works with any programming tool that supports the two-wire Spy-bi-Wire interface.

MSP430G2553. It's a low-power, 16-bit MSP430 microcontroller with an 8-channel, 10-bit analogue-to-digital converter (ADC), on-chip comparator, touch-sense-enabled inputs/outputs (I/Os), universal serial communication interface, 16kB flash memory and 512 bytes of RAM (preloaded with a sample program).

MSP430G2452. This too is a low-power, 16-bit MSP430 microcontroller with an 8-channel, 10-bit ADC, on-chip comparator, touch-sense-enabled I/Os and universal serial interface but it has an 8kB flash memory and 256 bytes of SRAM.

Microchip's chipKIT Uno32

The chipKIT platform from Microchip is a PIC32-based developer board that is natively compatible with the Arduino programming language and development environment. This board uses a modified version of the Arduino integrated development environment and existing Arduino resources like code examples, libraries, references and tutorials.

"Microchip's foundation started with the hobbyist community and we are also well entrenched in the



Freescale's i.MX53 Quick Start board

academic community worldwide. We were constantly receiving multiple requests and inquiries for a PIC MCU-based Arduino solution. So we decided to act on the voices we were hearing from these two groups and launched the PIC32-based chipKIT platform," said Cheri Keller, senior design partner program manager, Microchip Technology Inc.

These chipKIT boards are clocked

at 80 MHz, and provide 512kB Flash and 128kB RAM. The boards feature connectivity peripherals, including Ethernet, CAN and USB (full-speed host, device and OTG), plus peripherals such as multiple timers, a 16-channel 1MSPS ADC, two comparators, and multiple I²C, SPI and UART interfaces. In addition, the chipKIT platform features the MIPS32 M4K core from MIPS Technologies.

Design Professionals' Opinion on Various Development Boards

Amit Gohel, embedded design engineer: "MSP430 will be the most suitable board for beginners since it is very inexpensive and has free IDE, GRACE support, ultra-low-power core with good performance, mixed-signal MCU, and a variety of packages from DIP20 to LQFP. The best of all you get free samples from TI."

Binoy Johnson, hardware design engineer: "Arduino is best if you are designing controlling hardware, and is the most commonly used board for interactive projects. With the huge amount of support available everywhere, this would be the obvious solution for beginners like students and amateurs working on embedded projects. A lot of shields are also available for Arduino, which allows for easy stacking."

Shrenik Shikhare, embedded software engineer: "While the STM32 has comparatively limited information online, it is the lowest-cost solution out there, and so would be a better choice for cost-sensitive applications. On the other hand, the BeagleBoard is something that stands out among the rest. Its capabilities are much higher as it can port Linux and Android, and is a good choice for a beginner in embedded Linux. Of course, the Raspberry Pi is now in competition."

Vinay Chaddha, GVC Systems: "I am teaching programming to students of class IX to XII in Nithari Village using Raspberry Pi and students are just loving it. While developing commercial products, my favourite now is STM32, which can drive TFT and touchscreen display. I have discarded MSP430 and Arduino, as I don't find them useful for either commercial products or teaching now. In 2011, I had used Arduino for commercial designs as well as for trainings."

Zahir Parkar, firmware engineer: "MSP430 is the best choice when working with deeply embedded low-power products. For applications which demand higher processing power, 32-bit processors should be used. Here STM32 Cortex M3 comes to rescue. For even bigger projects or products that require a high-level operating system, Beagleboard can be used."